

Guanlin WU

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EDUCATION

Chinese University of Hong Kong, Shenzhen Computer Science and Engineering • MGPA: 3.470/4.0 • Core Courses: Computer Architecture, Operating System, Data Structure, Design and Analysis of Algorithm, • Optimization, Software Engineering	2020.09 – present Shenzhen China
UC Berkeley Academic Talent Development Program	2018.06-2018.08 Berkeley, CA

RESEARCH PROJECTS

University of California, Los Angeles/UAV Path Planning and Dynamic Obstacle Avoidance Research Assistant • Propose a new UAV system based on 2D-LiDAR that including the Map Construction (based on Hector SLAM), the Path Planner (using RRT) and the Dynamic Obstacle Handler (using D3QN). • Design algorithms for LiDAR detection (with noise decreasing and motion trajectory) and Reset in RL training. • Develop the new training strategy of RL model for Dynamic Obstacle Hanlder is directly trained in the ROS-Gazebo-PX4 simulator with UAV model “iris_rplidar”. Publication: Guanlin Wu, Zhuokai Zhao, and Yutao He. "RELAX: Reinforcment learning Enabled 2D-LiDAR Autonomous sytem for Parsimonious UAVs." Submitted.	2023.06 – Current Online
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SENIOR/AGV Resource Pool JAVA Engineer • Propose several technical trace to implement the AGV Resource Pool: KM Matching algorithm, Dijkstra with time window, Dynamic Merging. • Develop a simulator based on OpenTcs for the performance testing on various possible algorithms.	2023.02 – 2023.04 Beijing, China
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East China Normal University/Virtual Machine Allocation Research Assistant • Improved Best Fit Algorithm slightly for specific type of DVBP problem by using an innovative way of constructing the Balanced Binary Tree. • Managed to use Approximation algorithm to prove the ratio between modified Best Fit Algorithm and Optimal solution: $BFT(L) < \frac{20}{9} OPT(L)$ • Found the Virtual Machine data about their type from VMAgent and HUAWEI. Publication: Guo, Lifeng, Changhong Lu, and Guanlin Wu. "Approximation algorithms for a virtual machine allocation problem with finite types." <i>Information Processing Letters</i> 180 (2023): 106339.	2022.05 – 2022.11 Shanghai, China
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Tianjin Port/LiDAR-Based Hatch Localization Data Analyst • Used laser to collect information about shipment and stockpile and simultaneously created a 3D model using Point Cloud. • Applied Point Cloud and Matlab to complete the rasterization process, one of the three stages of accomplishing the hatch localization • After scanning a reference hatch with size 23.55*15.35m using LiDAR, the hatch localization algorithm was able to measure the hatch accurately with the same dimensions, satisfying the anti-collision requirement of error margins less than 0.5 meters. Publication: Jiang, Zeyi, Xuqing Liu, Mike Ma, Guanlin Wu, and Jay A. Farrell. "LiDAR-Based Hatch Localization." <i>Remote Sensing</i> 14, no. 20 (2022): 5069.	2021.07 – 2022.04 Tianjin, China
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COURSE PROJECTS

Operating System Course Projects: Independent Projects	2022.09 – 2022.12 CUHK(SZ)
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Learned about Kernel and loading a module into the Kernel

- Create a kernel thread and fork a process to execute the test program.

Used multithread programming to accomplish a game called “Frog crosses river” and implemented a thread pool

- This game is a multithread game, which requires forking threads for its implementation.
- Thread pool is one that acts a pool consisted with several threads, which will perform the job after a user requesting.

Implemented the simulation of a mechanism of virtual memory via GPU’s memory(CUDA Programming)

- Both physical memory and secondary memory has been implemented by using share memory and global memory in CUDA.
- Implement the memory replacement policy for page fault.
- Implement the map between virtual address(VA) to the physical address(PA).
- Implement a paging system with swapping between shared memory and global memory.

Implemented the simulation of a mechanism of file system management via GPU’s memory(CUDA Programming)

- Implement the logical file system that manages the metadata information, which includes all of the file-system structure except the actual data.
- Implement the free-space manager that tracks the unallocated blocks and provide them when needed.
- Implement a file-control block(FCB) which contains the information about the file, including ownership, permissions, and location of the file contents.

Design and Analysis of Algorithm Course Project: Pandemic Control Algorithm

2022.09 – 2022.12

Group Leader

CUHK(SZ)

- Implemented both the subtree-heavy algorithm and degree-heavy algorithm and conclude that subtree heavy performs 25% better than the other on average.
- Used Approximation algorithm to prove the ratio between Subtree-heavy Algorithm and Optimal solution, which is

$$S_{greedy} \geq \frac{1}{2} S_{optimal}$$

Database System Course Project: E-commerce Web Shopping Platform

2022.02 – 2022.05

Group Leader

CUHK(SZ)

- Developed a general understanding of the online shop platform.
- Utilized Java, Vue, Maven, and MySQL through the project.
- Implemented the CRUD operations through front-end with corresponding changes to back-end.
- Cultivated leadership and the ability of distributing tasks to members of the group.

Computer Architecture Course Projects

2022.01 – 2022.05

Independent Projects

CUHK(SZ)

Implemented a C++ translator for MIPS Instructions set to machine code

- This translator will scan the input file twice and return a lable table and a vector filled with each line of instructions.
- After processing, it will write the translated machine code into a .txt file.

Implemented a C++ simulator for the memory and register of the computer when running a MIPS asm file

- This simulator will diplay the change in Stack segment, Data segment(Dynamic data & Static data) and Text segment while running a MIPS code.

Implemented an Arithemetic and Logic Unit(ALU) by using Verilog.

- This program implements a simple CPU which supports simple instruction parsing, Register Value fetching, and ALU functions such as add, addi, etc.

Implemented a 5-stage pipelined CPU by using Verilog

- This program mainly contains 5 clock cycle to execute one instruction.
- In the first clock cycle, the CPU read one instruction in the instruction memory with the newly given pc address.
- In the second one, the processor will divide the instruction to different parts and decode the MIPS instruction with the registers and control unit.
- In the third one, ALU will handle arithmetic, logical, shifting and conditional branch instructions.
- In the fourth one, data will be fetched from or stored to the data memory by data transfer instructions.
- In the fifth one, data will be written back to registers if needed.

SKILLS&INTERESTS

Language: English (TOEFL 108; GRE 326+4), Chinese (native)

Computer skills: C, C++, Java, Python, mysql, Matlab, Point Cloud, Verilog, CUDA programming

Interests: Philosophy, RAP, Work out, Basketball